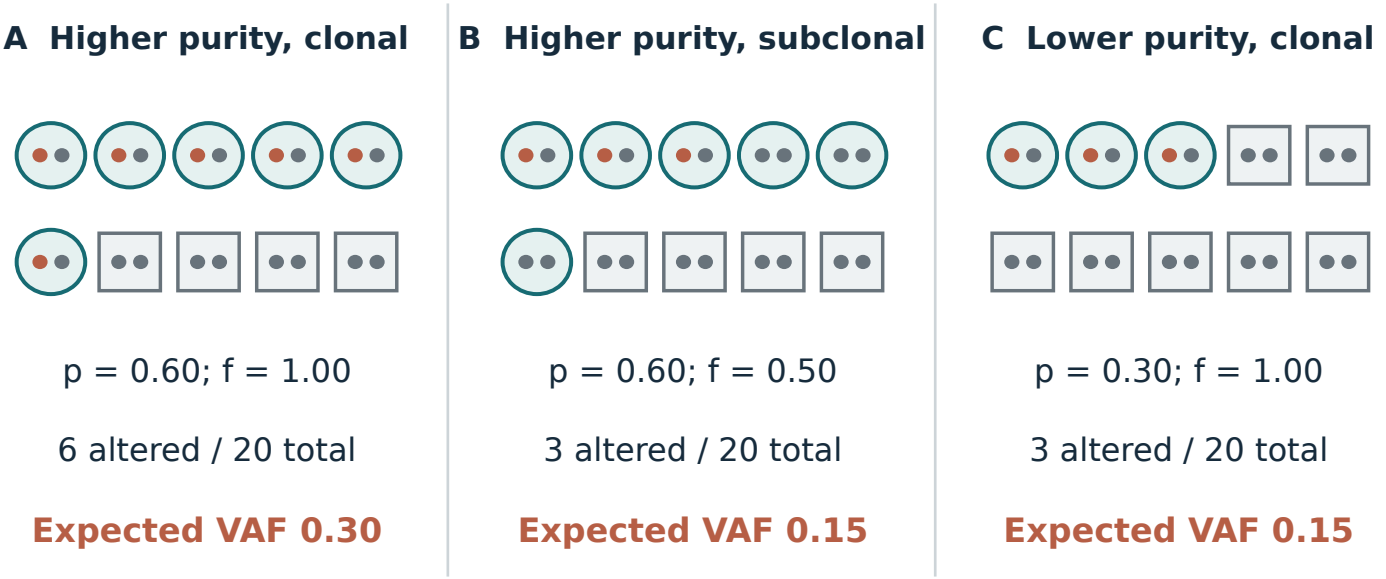
The same allele fraction can describe different mixtures

Invented ten-cell samples; two local DNA copies per cell



B and C both have 3 altered DNA copies among 20 total copies.

Different tumour purity and cellular prevalence can produce the same expected VAF.

Tumour cell Normal cell Altered copy

Normal cells carry no altered copy. Counts are an idealized expectation, not observed reads; equal sampling and no copy-number change are assumed.